

Discerning Deepfakes Using Low-Level Spatio-Temporal Features

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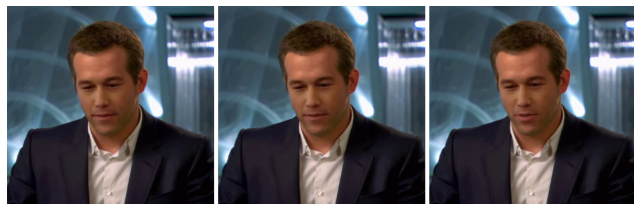
Abstract

Deep Generative modeling has taken giant leaps in performance in the recent years. It has now become easier than ever to generate synthetic media that is nearly indistinguishable from the real media. This, however, also has a negative side: DeepFakes. With easy access to technology to create DeepFakes, malicious uses of this technology have been increasing. DeepFake videos are becoming more common on social media platforms everyday. In order to counter this, a lot of research effort is being put into creating DeepFake detection methods. Some deep learning models have shown good performance and robustness in this task. However, most of these are mammoth models with millions and millions of parameters. Such models have huge computational costs and can be hardly used for fast inference in real-world scenarios. Additionally, most of these methods work with images and discard important temporal information in the videos. This paper proposes two lightweight DeepFake detection models which uses Recurrent Convolutional layers to aggregate features across the time dimension: Micro-CRL and Meso-CRL with merely 6,809 and 14,371 parameters, respectively. Experiments show that the proposed models can efficiently use the low level spatio-temporal information in videos to detect DeepFakes, achieving the state-of-the-art results on the publicly available benchmark DeepFake datasets.

1. Introduction

Fake images and videos are not something new, facial manipulation has been a topic of interest since the 1990s [1] and recent advances in deep learning have given a boost to this field. Transitioning from the primitive face-swapping technology [2], we now have DeepFakes [3, 4, 5]. “Deep-Fake” is an umbrella term used to refer to deep learning based techniques used to create fake images by swapping the face of a person by the face of another person. Typically,

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(a) Identity of Ryan Reynolds swapped with Joseph Gordon-Levitt



(b) Identity of Ryan Reynolds swapped with Randall Park

Figure 1. These are the samples from Celeb-DF Dataset. There are no visible artifacts in the frames, frames are temporally coherent, original expressions are preserved and illumination is consistent with the rest of the scene.

synthetic images and videos are created using autoencoders (AE) [6] and Generative Adversarial Networks (GANs) [7], and the product of these deep learning algorithms is almost indistinguishable to the naked eye [6]. Fig. 1 shows how realistic Deepfakes have become.

This innovation in the field of computer vision has recently started to be abused by people with malicious intentions who have been generating fake videos of female celebrities [8] and the general public. Deepfakes, in fact, have gone from being an innovative harmless tool to a perfect tool for somebody seeking to exert control and power over a victim. Photo-shopped revenge porn existed before but it was not very common, however, the ease of availability of tools to create DeepFakes [9, 10, 11] is making this a common practice. There were around 7,964 Deepfake videos online in December 2018 and in the following nine months the number almost doubled to 14,678 [12]. Furthermore, the malicious uses of deepfakes are not limited to porn only. Deepfake technology can become a power-

ful tool in spreading politically motivated lies. Fake videos can be quickly created using this technology and, using the power of social media, such videos can be spread worldwide almost instantly. In May 2019, a deepfake video of Donald Trump surfaced in which he offered advice to Belgium people on climate change and this sparked an outrage in Belgium where people viewed this as an intervention of the United States of America in Belgium’s climate policy [13]. President Donald J. Trump also shared a video of his political rival, Nancy Pelosi, in which she appeared to be intoxicated, and this video was later widely shared on Facebook and Youtube [14].

In response to these continuously evolving deepfake generation methods, several deepfake detection approaches [15, 16] have been proposed to detect such forgery content. Some of the approaches use visual inconsistencies present in the videos, such as eye blinking rate, unnatural head poses and illumination [17, 18, 19, 20], to detect the deepfakes while others are end-to-end deep learning models. Generally, the end-to-end deep learning based detection models are more robust and efficient than methods using feature engineering [15]. The deep learning based detection methods are further generally divided into two types: methods based on spatial information (single frame): [21, 22, 23] and methods based on spatial and temporal information (sequence of frames) [24, 25]. In addition, several datasets have also been made publicly available, such as Celeb-DF [26], FaceForensics++ [27], UADFV [18] and Deeper Forensics [28]. The UADFV and FaceForensics++ are commonly considered first generation Deepfake datasets while Celeb-DF and Deeper Forensics are referred to as second generation Deepfake datasets. The quantity, quality, and variation in the Deepfake videos in the second generation datasets is much higher as compared to the first generation datasets [15], and therefore these are considered more challenging. All of these datasets contain both the real and manipulated videos, with the real videos mostly collected from online video streaming sites like YouTube while Deepfake videos are generated using different techniques. This paper utilizes publicly available benchmark datasets from both the generations to analyse generalization and robustness of the proposed models (Micro-CRNN, Meso-CRNN) for deep fake detection. The results indicate supremacy of the proposed models with respect to the state-of-the-art, where the current paper in particular makes the following contributions.

- Presents a simple Convolutional Recurrent layer that can efficiently capture spatio-temporal features.
- Proposes two light-weight Deepfake detection models, that detect the Deepfakes using low level spatial-temporal features.

- Outperforms the state-of-the-art DeepFake detection approaches on several recent DeepFake datasets.

2. Related Work

2.1. Deepfake Generation

Facial manipulation techniques are not new, they have been around since 1990s. Bregler *et al.* [29] presented a technique to generate mouth movements so a person can be seen saying the words that he/she did not say. Dale *et al.* [30] presented an automatic face replacement technique. They tracked the 3D facial movement in both source and target videos and exploit the corresponding 3D geometry to warp the source face to the target face. Garrido *et al.* [31] presented a system that replaces the face of an actor with a target while preserving the original expressions. This technique was able to work with limited data. Thies *et al.* [32] demonstrated the first real-time expression transfer for facial reenactment. In 2017 the faceswapping technique emerged that used a pair of autoencoder networks [6] with a shared encoder [10]. It was named Deepfake after the username of Reddit user that used this technique to generate face-swapped adult content. However, the autoencoders failed to reconstruct fine details in the images [6] and the model had to be retrained for each pair of identities [10].

The improvements of Generative Adversarial Networks (GANs) [7], and its variants [33], have made it possible to create high resolution realistic fake images and videos that are nearly impossible to distinguish from the real ones. Liu *et al.* [34] proposed an unsupervised image to image translation framework based on coupled GANs, which aims to learn the joint representation of images in different domains. This algorithm became the basis for generating the high quality Deepfake content. Zhu et al. [35] proposed Cycle-GAN to utilize cycle-consistent loss. The idea was to demonstrate that image translation from one domain to another is reversible. Bansal et al. [36] advanced the work and proposed Recycle-GAN, which incorporated spatial and temporal information with conditional generative adversarial networks. StarGAN [37] learned the image-to-image translations across multiple domains using a single model.

2.2. DeepFake Detection

2.2.1 DeepFake detection using single image

There are various techniques in field of multimedia forensic for detection of forgeries and alterations in single images. Various detection methods using simple features have been proposed. Ricard *et al.* [38] applied Discrete Fourier Transform and Azimuthal Average to create 1D representation of the FFT power spectrum and fed it to logistic regression, Support Vector Machine (SVM), and K-Means Clustering to discriminate between altered and pristine images. Xin

et al. [18] estimated the 3D head poses from images and trained an SVM classifier on them to distinguish real videos from fake images.

The detection of deepfakes using classical techniques, which are dependent on hand-crafted features, is gradually becoming obsolete [39], due to the rapid improvements in the deepfake generation techniques. Whereas, the deep learning based approaches are considerably taking the mainstream [15] for the deepfake detection task.

The target of many deep learning based detection methods is to find the subtle artifacts in the deepfake videos which often evade human perception. Yuzezun *et al.* [23] proposed to detect the artifacts in the face and its surrounding regions using CNN based detection mechanism. Four different CNNs were trained, to detect fake images, from scratch: VGG16 [40], ResNet50, ResNet101, and ResNet152 [41]. The proposed detection method achieved state-of-the-art results for UADFV [18] and Deepfake-TIMIT [39] datasets. Afchar *et al.* [21] demonstrated that it is difficult to identify forged images at high semantic level especially if they contain human faces, and the noise in images at microscopic level makes analysis difficult therefore the features at mesoscopic level can be used to detect Deepfake images. [21] introduced two CNN networks composed of a few layers: Meso-4 and MesoInception-4 that used the mesoscopic features of the images to classify Deepfake images. The proposed architectures achieved high detection rate of 98% on q private Deepfake dataset and 95% for FaceForensic++ Deepfake dataset [15]. Zhou *et al.* [22] proposed a two-stream network for face manipulation detection: i) a face classification stream based on the GoogLeNet [42] to detect fake facial image, and ii) a path triplet stream that was trained with a triplet loss using steganalysis features of image patches, and an SVM for the classification. Nguyen *et al.* [43] proposed the use of capsule networks for detecting fake media. The output of the three capsules is routed to the output capsules using a dynamic routing algorithm. It is done for several iterations to distinguish between fake and real media. Rössler *et al.* [27] did exhaustive analysis of Deepfake detection mechanism. The XceptionNet using ImageNet pretrained weights [44] performed the best when retrained for the Deepfake classification task. Zhao *et al.* [45] recently proposed a patch-wise consistency learning (PCL), and have achieved AUC score of 99.79% on FF++ dataset. Recently, Hong *et al.* [46] proposed a lightweight non-DL Deepfake detector: DEFAKE-HOP. The proposed model had 42,845 parameters and was able to outperform DL approaches on the CelebDF [26] dataset.

The still frame approaches discard the important temporal information from the videos and therefore are sub-optimal.

2.2.2 Deepfake Detection using multiple frames

The DeepFake videos can be generated by manipulating still frames and stitching the manipulated frames together along the temporal axis. Most DeepFake video detection techniques employ the strategy of aggregating the per frame classification score to obtain the classification of the entire video. This frame-wise analysis of videos ignores the temporal features in the videos.

Güera *et al.* [25] argued that the way the Deepfakes are generated, intraframe or temporal inconsistencies emerge between frames. These inconsistencies could be used in detection. The authors also proposed a CNN-RNN architecture to detect the Deepfakes. First spatial features are extracted from 'n' consecutive frames using CNNs, then these extracted features are passed to LSTMs to create a temporal sequence descriptor. Finally, a fully connected layer is used for making the prediction. By combining recurrent network such as LSTM with CNN, the resulting networks are considered to be 'deep in space' and 'deep in time' and thus are well suited to exploit the intra-frame inconsistencies. Sabir *et al.* [24] proposed a similar recurrent convolutional model based on the combination of the DenseNet CNN [47] and the Gated Recurrent Unit (GRU) cells [48] to exploit the across-frame discrepancies.

Shi *et al.* [49] argued that using the FC-LSTM to handle the spatio-temporal data is sub-optimal due to the full connections during input-to-state and state-to-state transitions where no spatial information is encoded. Therefore, the CNN-RNN models proposed by Guera *et al.* [25] and Sabir *et al.* [24] are not very efficient at detecting across-frame discrepancies, especially in newer DeepFake datasets

2.3. Recurrent Convolutions

In order to overcome the short-comings of a standard FC-LSTM and preserve spatial information Shi *et al.* [49] presented Convolutional LSTM layer (ConvLSTM). ConvLSTM has convolution operations in the state-to-state and input-to-output operations. The following equations represent the operation of ConvLSTM layers. (The Hadamard product and the Convolution operator are denoted by 'o' and '·', respectively).

$$i_t = \sigma(W_i X_t + U_i H_{t1} + V_i \odot C_{t1} + b_i)$$

$$f_t = \sigma(W_f X_t + U_f H_{t1} + V_f \odot C_{t1} + b_f)$$

$$C_t = f_t \odot C_{t1} + i_t \odot \tanh(W_c X_t + U_c H_{t1} + b_c)$$

$$o_t = \sigma(W_o X_t + U_o H_{t1} + V_o \odot C_t + b_o)$$

$$H_t = o_t \odot \tanh(C_t)$$

ConvLSTM and its variants have been successfully used for different video based tasks e.g. video action recognition

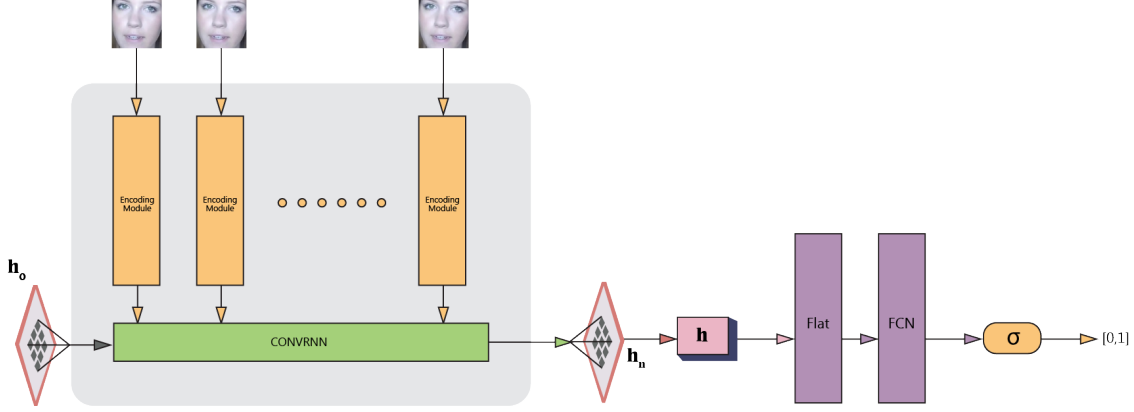


Figure 2. Proposed System Architecture. Weights are shared between the encoding modules. In Micro-CRL and Meso-CRL only the encoding modules are swapped.

[50], video forecasting [51], video super resolution [52], video object detection [53], gesture detection [54] and optical flow estimation [55].

Shahroz *et al.* [56] used the ConvLSTM cell proposed by Shi *et al.* and constructed a deep residual network which takes, as an input, a sequence of consecutive frames from a video and detect unnatural looking artifacts in spatio-temporal domain. They performed their experiments on FaceForensics++ dataset to show the robustness and generalization of their approach. However their good performance comes at the cost of efficiency, as the proposed model is a mammoth model with 250 million parameters which limits its use in the real world scenarios where the inference time and computation costs are an important factor. The methods proposed in the paper address these issues without compromising too much on the detection performance.

3. Methodology

The creation of Deepfakes usually introduces several artifacts in the video frames. Some of these artifacts are visible to human eyes and thus can be detected easily [23]. However, most artifacts in the recent Deepfake videos are not very prominent and exist as low level noise in the videos, such as inconsistent facial texture, unnatural changes in illumination of the parts of the face, blurriness around eyes and mouth and flickering across facial boundaries which are visually hard to detect. However, by analysing microscopic features such as edges and object boundaries and mesoscopic features such as object shapes, sizes and color gradients across a short sequence of video frames, subtle spatio-temporal changes can be detected.

The analysis of low-level image features comes with a huge challenge of dealing with noise. Video compression, format conversion and even minor video processing greatly change the microscopic and mesoscopic features of video

frames. Any method used for the analysis of microscopic and mesoscopic features needs to be robust to the noise. The methods based on manual feature extraction [38] are not very robust to the noise and perform poorly [8]. The deep learning methods are robust to the noise and can extract better features than the handcrafted feature methods.

Doing low-level image analysis has an added advantage of reduced complexity. It is not difficult to extract low-level image features therefore, a small and simple CNN can be used for feature extraction.

In this regard two lightweight models are proposed: Meso-CRL and Micro-CRL. The proposed models consist of three parts: a backbone model consisting of Inception modules and/or convolution layers, a sequential module consisting of Convolutional Recurrent Layer(s) and a Classifier module consisting of Fully Connected layers.

Figure 2 represents the general architecture of the proposed models. The backbone of both models is a shallow and lean Convolutional Neural Network. The backbone acts as a feature extractor for the sequence of images, the output of the backbone is aggregated across time using a simple Convolutional Recurrent Layer (CRL). CRL is a variant of ConvLSTM [49]. The output of the CRL is passed through global average pooling layer and flattened. The flattened vector is then passed through two Fully Connected layers to get the prediction for the sequence. The following section describe each module of the architecture in detail.

3.1. Convolutional Recurrent Layer

Convolutional recurrent neural networks (CRNNs) are the convolutional version of RNNs, designed to represent spatio-temporal data. CRNN differ from standard RNN in that it has convolution operation in both the input-to-state and state-to-state transitions. The CRNN determines the future state of a certain point in the grid by the current inputs and the past states of its local neighbors. This can be achieved by using a convolution operator in the input-to-

state and state-to-state transitions. Fig. 3

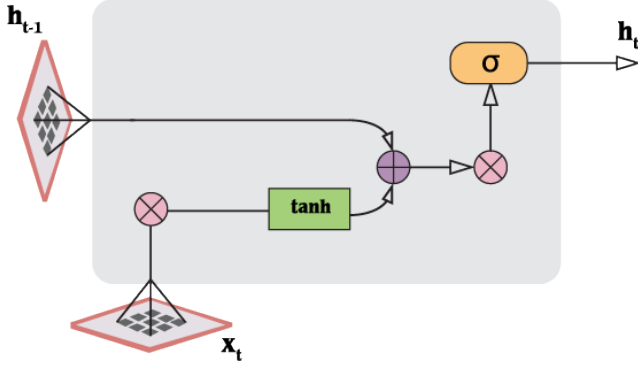


Figure 3. Proposed ConvRNN cell

The 2D input feature vector X_t is passed through a 2D spatial convolution layer, which maps X_t to i_t . i_t has the same number of channels as the hidden state of the CRL. The $\tanh$ activation is then applied to i_t , and the resulting vector is combined with previous cell state H_{t-1} via addition operation to get f_t . f_t is then passed through a second 2D convolution layer to get the new cell state H_t . The equations below summarises the operation of Convolutional Recurrent cell. ‘*’ denotes the convolution operator:

$$i_t = \tanh(W_{xi}X_t + b_i)$$

$$f_t = (i_t + H_{t-1})$$

$$H_t = \sigma(W_{fh}f_t + b_h)$$

CRL uses two convolution operations and has 4x less parameters than ConvLSTM. The proposed Deepfake detection method involves spatio-temporal analysis across a short sequence of frames (8-10 frames), therefore a simpler CRL suffices. To demonstrate this empirically, experiments have been performed to establish that ConvLSTM provide no added advantage at the cost of increased number of parameters.

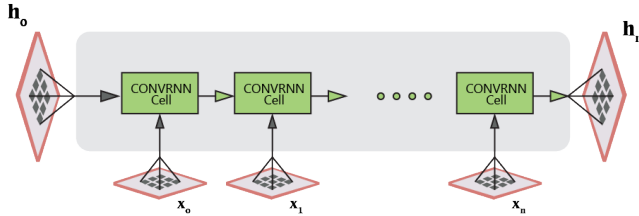


Figure 4. Unrolled ConvRNN layer

3.2. Micro-CRL model

The Micro-CRL is a shallower version of the proposed model, consisting of 1x 5x5 convolution and 2x 3x3 convolutions. The size of the input frames to the model is 64x64

and 5x5 convolution in the beginning is used to capture any long term relations between the pixels. All Convolution layers in the backbone are followed by Batch Normalization and Max Pooling layers. The CRL consists of 16x 3x3 convolutions to capture the local spatio-temporal features.

The total number of trainable parameters for this model is 6,809K. For comparison it is approximately 3700 times smaller than the standard ResNet50 [41] model and 2 times smaller than the most lightweight model it is evaluated against 6. Table 1 gives further details about the number of parameters in each layer of the model, and Figure 5 visually represents the backbone CNN of this variant.

The backbone of the Micro-CRL is intentionally kept lean and simple to reduce the complexity of the model, so that it is only able to capture very simple micro-scopic features from the videos.

Layer	Total Parameters
4x 5x5 Convolution	320
8x 3x3 Convolution	328
16x 3x3 Convolution	1232
16x 3x3 CRL	4640
16x Fully Connected	272
1x Fully Connected	17

Table 1. Number of parameters in each layer in Micro-CRL

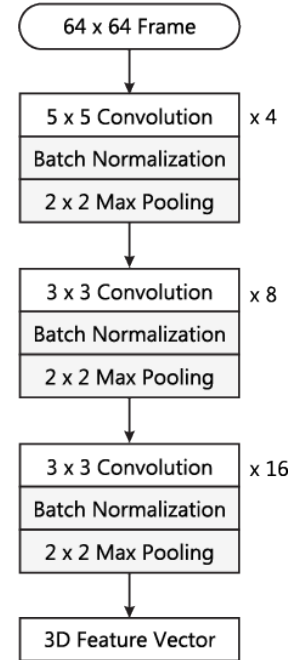


Figure 5. Micro Inception Encoding module

3.3. Meso-CRL model

The backbone CNN of this variant consists of three Inception modules followed by Batch Normalization and Max

pooling and two 5×5 Convolution layers with Batch Normalization. Similar to Micro-CRL this models also contains CRL with $16 \times 3 \times 3$ convolutions to capture local spatio-temporal features.

The original variant of the inception module [57] consists of a path with 5×5 convolutions. This layer was replaced with 3×3 dilated convolutions [58] to reduce high semantics and the number of parameters, while keeping the same receptive field. This variation is presented by Afchar *et al.* [21]. The chosen parameters (ai, bi, ci, di) for the Inception module 'i' can be found in Table 2.

The total number of parameters for this model is 14, 371. Meso-CRL is twice the size of Micro-CRL, this increase in the number of parameters is mainly due the increase in the depth of the model. The depth is still very shallow as compared to standard CNN models: GoogleNet 22 layers deep, Resnet-50 and Resnet-152 are 50 and 152 layers deep, respectively. The shallow backbone of Meso-CRL limits the features that it can capture to meso-scopic levels.

Table 3 present more details about the total number of parameters in each part of the model and figure 6 visually represents the backbone CNN of Meso-CRL model.

Layer	a	b	c	d
1	1	4	4	1
2	2	4	4	2
4	4	8	8	4

Table 2. Number of filters for each convolution in Inception modules used in Meso-CRL

Layer	Total Parameters
8x 5×5 Convolution	640
Inception Module	436
Inception Module	514
16x 5×5 Convolution	4880
Inception Module	1820
16x 3×3 CRL	5792
16x Fully Connected	272
1x Fully Connected	17

Table 3. Number of parameters in each layer in Meso-CRL

4. Experiments

Extensive experiments on different datasets are conducted to show the effectiveness and robustness of the proposed models.

4.1. Datasets

To test the performance of the proposed method, 4 different first and second generation DeepFake datasets were selected: UADFV [18] FaceForensics++ [27], Celeb-DF v2 [26] and Deeper Forensics 1.0 [28]. The current paper

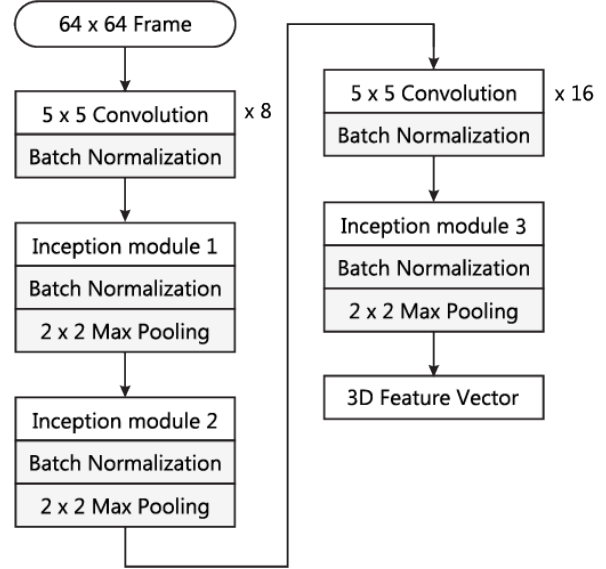


Figure 6. Meso Inception Encoding module

also presents a comparative analysis of the proposed models against the state-of-the-art methods on the above mentioned datasets.

4.1.1 UADFV

UADFV [18] is a first-generation released in 2018. It is small Deepfake dataset with just 98 videos of which 49 are real and 49 are fake. Each video has one subject and lasts approximate 11.4 seconds. The real and fake videos were randomly split into training and testing sets with 80% and 20% ratios respectively.

4.1.2 FaceForensics++

FaceForensics++ [27] is another first generation Deepfake dataset released in 2019. It consists of 1000 original videos that have been manipulated using four different techniques: Deepfake (DF), Face swap (FS), Face2Face (F2F) and Neural textures (NT). The data has been sourced from 977 YouTube videos. The authors also provided binary masks which can be used to segment the facial regions. In all the experiments performed on this data, the official training, validation, and testing sets were used. Training, validation and testing sets consisted of 1440, 280, and 280 videos respectively. The real and Deepfake videos were available at 1 : 1 ratio.

4.1.3 Celeb-DF v2

The aim of the Celeb-DF v2 [26] dataset was to provide large number of fake videos of better visual quality compared with the previous datasets. [26]. This dataset consists

of 590 real videos extracted from YouTube, corresponding to celebrities with a diverse distribution in terms of gender, age, and ethnic group. In addition there are 300 random videos from YouTube with varying subjects. There is great variety in videos regarding illumination, pose and distance of the subject from the camera. Regarding fake videos, a total of 5,639 videos were created swapping faces using DeepFake technology. Average length of each video is 13 seconds. There are approximately 225.4K frames in the real videos and 2116.8K frames in the fake videos. For training and testing the original benchmark splits provided by the authors were used.

4.1.4 DeeperForensics (DFR) v1.0

DeeperForensics v1 [28] is a recent large scale dataset. Authors collected 100 identities using paid actors and swapped each identity onto target videos collected by FaceForensics++ [27]. After swapping the identities 7 different perturbations, like color saturation, blurring and compression, different parameters were applied to the output so as to better represent a real-world scenario. The dataset contains 10,000 fake videos and 1000 real videos. There are approximately 5 million fake frames and 500K real frames. The data is split into training, validation and testing in the ratio 0.7:0.1:0.2 respectively, using the splits provided by the authors.

4.2. Face Image Preprocessing

The videos are pre-processed to extract the facial region. Two different face detectors: Dlib [59] face detector and MTCNN [60] were used for this purpose. The Dlib face detector is fast but the detection accuracy is low, on the other hand, the MTCNN face detector is slow but has a higher accuracy. So to speed up the pre-processing steps and get high detection rates, DLib face detector and MTCNN were used in sequence: False Negative frames by DLib were processed using MTCNN. The facial-images were resized to 64×64 to reduce the complexity of the input space. The final processing step involved image mean normalization and re-scaling from 0 – 1 before feeding them to the network.

4.3. Detection performance

Extensive experiments were performed to judge the performance of the proposed methods on the Deepfake detection task. The DeepFake datasets: Celeb-DF and Deeper Forensics have imbalanced class distribution and have a higher ratio of fake videos as compared to real videos. So, for the evaluation of the models, Area under Receiver operating curve (AUC-ROC) metric is used. Table 4 summarizes the video-level performance of both Micro-CRL and Meso-CRL architectures, in terms of AUC score, on different datasets.

Table 4. Detection performance measure in terms of AUC score on different datasets

Dataset	Micro-CRL	Meso-CRL
UADFV	95.45	100
FF++ (RAW)	96.71	99.64
Celeb-DF v2	92.26	94.12
DFR v1	98.60	98.92

The proposed models show good performance on both generation *I* (UADFV, FF++) and generation *II* (Celeb-DF, DFR) datasets. Meso-CRL shows better performance than Micro-CRL on all the datasets. This was expected due the greater noise in the Micro-scopic features as compared to the Meso-scopic features. On the DFR dataset, difference in performance of the two methods is relatively small, confirming that the videos in DFR are of high quality with little low level noise. Meso-CRL achieves perfect 100% AUC score on UADFV dataset despite this dataset having very few (78) training videos. This indicates that the proposed model can learn generalized features from a small number of samples.

Video are often compressed, for example while uploading on a social media website, in the real world scenarios. This compression usually reduces the video quality and also introduces several artifacts. To emulate the real-world compression, FF++ provide videos with two different compression types c23 (high quality videos with quantization parameter equal to 23) and c40 using a quantization of 40.

Most of the artifacts introduced by the video compression are at Micro and Meso scopic levels so they can severely effect the performance of the proposed models. Experiments on the compressed FF++ videos were performed to test the robustness of the proposed models against video compression. Table 5 summarises the results of these experiments. The performance of both models suffer almost similarly as the quantization rate increases, but they still show high detection rates.

Table 5. Detection performance on FaceForensics++ dataset with different levels of compression

Compression	Micro-CRL	Meso-CRL
c23 (HQ)	93.50	94.64
c40 (LQ)	90.67	93.39

4.4. CRL vs ConvLSTM

Experiments on FaceForensics++ (FF++) and CelebDF were conducted to compare the performance of the proposed variant of Recurrent convolutional layer (CRL) against ConvLSTM layer [49]. Meso-CRL is compared with Meso-ConvLSTM. The only difference between the

Table 6. Comparison of the size and the detection performance of benchmarking methods. Video-level AUC score is considered as the evaluation metric. In the case of AUC performance bold face and underline indicates the best and the second-best results, respectively. In the case of Number of parameters bold face and underline indicates the smallest and the second-smallest model, respectively. Most AUC results of benchmarking methods are taken from [26] and [15]. The information about number of parameters for most methods is taken from [46]

	Method	Celeb-DF v2	FF++/DF	UADFV	Number of parameters
Zhou <i>et al.</i> [22]	InceptionV3	53.8	70.1	85.1	24M
Afchar <i>et al.</i> [21]	Meso4	54.8	84.7	84.3	27.9K
Afchar <i>et al.</i> [21]	MesoInception-4	53.6	83.0	-	28.6K
Li <i>et al.</i> [23]	FWA(ResNet-50)	56.9	80.1	97.4	23.8M
Yang <i>et al.</i> [18]	HeadPose(SVM)	54.6	47.3	89	-
Matern <i>et al.</i> [20]	VA-MLP	53.8	66.4	70.2	-
Rossler <i>et al.</i> [27]	Xception-raw	48.2	99.7	80.4	22.8M
Nguyen <i>et al.</i> [61]	Multi-task	54.3	76.3	65.8	-
Nguyen <i>et al.</i> [43]	CapsuleNet	57.5	96.6	61.3	3.9M
Sabir <i>et al.</i> [24]	<i>DenseNet+RNN</i>	79.1	<u>99.6</u>	-	25.6M
Li <i>et al.</i> [23]	DSP-FWA (SPPNet)	64.6	93	<u>97.7</u>	-
Tolosana <i>et al.</i> , [62]	<i>Xception</i>	83.6	99.4	100	22.8M
Hong <i>et al.</i> [46]	DeepfakeHop	90.56	97.45	100	42.8K
Shahroz <i>et al.</i> [56]	CLRNet	-	99.02	-	250M
Ours	Micro-CRL	<u>92.26</u>	93.50	95.45	6.7K
Ours	Meso-CRL	94.12	94.60	100	<u>10K</u>

two models is the Recurrent convolutional layer. The number of filter and the shape of filters in each layer are kept the same. ConvLSTM layer with 16x 3x3 filters has 16192 parameters and CRL layer with 16x 3x3 filters has only 4640 parameters. Table 7 show the AUC scores obtained from both models on each dataset. ConvLSTM with 4x parameters offer no significant advantage over CRL. Furthermore, the generalization of Meso-ConvLSTM model suffer when the size of the training set reduces.

Table 7. CRL vs ConvLSTM

Dataset	Meso-CRL	Meso-ConvLSTM
(FF++) HQ	94.60	90.03
(FF++) LQ	93.39	89.8
Celeb-DF	94.12	96.0
Avg	94.03	91.94

4.5. Micro-CRL vs Handcrafted features

Compression of the videos introduce many low-level artifacts in the videos which can effect any method which is using those features for classification. Experiments are done, on FF++ dataset with high-compression rate (c40), to see how Micro-CRL handles the noise at the low level as compared to a method based on low-level hand-crafted fea-

ture: power-spectrum analysis [38], dynamic-texture analysis [63] and Visual-artifacts in eyes and teeth [64]. Table 8 shows the results of the experiments.

Experiments show that methods based on low-level hand-crafted features perform poorly as compared to Micro-CRL when evaluated on videos with high compression rate. This is because low-level noise, introduced during video compression, disturb the image features that these methods were using for the classification. Micro-CRL, on the other hand, learns more effective features. The performance of Micro-CRL on FF++(LQ) show that it is much more robust to noise as compared to hand-crafted techniques.

Table 8. Performance comparison of Micro-CRL and methods based on handcrafted features on FF++(LQ) dataset

Method	AUC
Micro-CRL	90.67
Power spectrum analysis [38]	62.8
Visual artifacts [64]	-
Dynamic texture analysis [63]	80.49

4.6. Comparison with Other Methods

A comparative analysis against the state-of-the-art techniques is performed to show the effectiveness of the pro-

posed approach 6 The results show that all Deep Learning based methods struggle on the Celeb-DF v2 dataset. This is because the Deepfake videos in CelebDF are of higher quality as compared to other datasets. Many Deep Learning based Deepfake approaches detect the artifacts that may be introduced while splicing the synthesized face into the original frame [23], [27], [43]. In the Celeb-DF Deepfake videos several post-processing steps were taken to remove artifacts introduced during swapping process, as a result performance of many Deepfake detectors dropped. A recent non-DL method, by Hong *et al.* [46], performs reasonably well on the Celeb-DF dataset. Please note that both proposed models: Micro-CRL and Meso-CRL outperform all the DL and non-DL based methods clearly on Celeb-DF v2. Furthermore, the proposed Meso-CRL achieves the state-of-the-art and near state-of-the-art performance on UADFV and FF++ datasets. The Micro-CRL does not lag behind much. Despite having the lowest number of parameters (only 6.7K against millions for the state-of-the-art), it gives the second best performance on Celeb-DF (only after the proposed Meso-CRL) and nearly state-of-the-art comparable performance on the FF++ and UADFV datasets.

5. Conclusion

Although the current Deepfakes have become visually stunning they still have many artifacts at Meso and Microscopic levels and this paper shows that the proposed Deep Learning models can be used to detect those artifacts. The added advantage of detecting the Deepfakes using low level features is that the detectors can be made small and simple. The experiments across different datasets show that the proposed detectors: Micro-CRL and Meso-CRL can efficiently detect the Deepfakes using low level spatio-temporal features. Additional experiments show robustness of the models against low-level noise. Finally a comparison with state-of-the-art methods show that the proposed models (with only a few thousand parameters) can compete with, and in some cases beat the, mammoth DL-models having millions of parameters.

The proposed models perform good on different Deepfake datasets, however, a further study needs to be done to check how they perform in the *wild*. Furthermore, research effort need to be done to make the proposed models further generalized across different datasets.

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